# Piper PA-28-161 Warrior II/III (PA28-161)

Lycoming O-320-D3G, 160 hp · 4 seats · Typical Warrior II/III (1977+) POH procedures; carbureted engine, manual flaps, 2,440 lb MTOW. Warrior III figures identical unless noted.

## V-speeds & limits

| | |
|---|---|
| Vr | **50–60 KIAS (45–55 KIAS short field, flaps 25°)** |
| Vx | **63 KIAS** |
| Vy | **79 KIAS** |
| Best glide | **73 KIAS** |
| Vfe | **103 KIAS** |
| Va | **111 KIAS (2,440 lb) / 88 KIAS (1,531 lb)** |
| Vno | **126 KIAS** |
| Vne | **160 KIAS** |
| Vs / Vso | **50 / 44 KIAS** |
| Approach | **63 KIAS (flaps 40°)** |

## Normal procedures

### Before Start

| | |
|---|---|
| Preflight inspection | **COMPLETE** |
| Fuel quantity | **CHECK visually (25 gal usable per tank, 48 total)** |
| Baggage door | **CLOSED & LOCKED** |
| Passenger brief | **COMPLETE (belts, door, exits, sterile cockpit)** |
| Seats, belts, harnesses | **ADJUST & LOCK** |
| Door | **CLOSED & LATCHED (top and side)** |
| Parking brake | **SET** |
| Avionics master | **OFF** |
| Electrical switches | **OFF** |
| Circuit breakers | **CHECK IN** |
| Alternate static source | **OFF (closed)** |
| Fuel selector | **FULLER TANK** |
| Carb heat | **OFF (full forward)** |

### Engine Start

| | |
|---|---|
| Mixture | **FULL RICH** |
| Primer | **PRIME 0–3 strokes (as required), LOCKED** |
| Throttle | **OPEN 1/4 inch** |
| Master switch | **ON** |
| Electric fuel pump | **ON** |
| Beacon | **ON** |
| Propeller area | **CLEAR** |
| Starter | **ENGAGE (10 s max, then 30 s cool-down)** |
| Throttle | **ADJUST 800–1,000 RPM** |
| Oil pressure | **CHECK rising within 30 s (60–90 psi green)** |
| Ammeter / alternator | **CHECK charging** |
| Electric fuel pump | **OFF (fuel pressure check 0.5–8 psi)** |
| Avionics master | **ON** |
| Flaps | **RETRACT (check handle latches)** |

### Before Taxi

| | |
|---|---|
| Flight instruments / avionics | **CHECK & SET** |
| Altimeter | **SET** |
| Heading indicator | **SET to compass** |
| Transponder | **STBY / as required** |
| Parking brake | **RELEASE** |
| Brakes | **CHECK during initial roll** |
| Steering | **CHECK (nosewheel steering with rudder pedals)** |
| Gyros | **CHECK during turns (turn coordinator, DG, attitude)** |

### Before Takeoff

| | |
|---|---|
| Parking brake | **SET** |
| Flight controls | **FREE & CORRECT** |
| Fuel selector | **FULLER TANK** |
| Mixture | **FULL RICH** |
| Throttle | **2,000 RPM** |
| Magnetos | **CHECK (max drop 175 RPM, max difference 50 RPM)** |
| Carb heat | **CHECK (RPM drop), then OFF** |
| Vacuum | **4.8–5.2 in Hg** |
| Oil pressure / temperature | **CHECK GREEN (60–90 psi)** |
| Ammeter | **CHECK charging** |
| Annunciator panel | **PRESS TO TEST (if fitted)** |
| Throttle | **IDLE check (500–700 RPM), then 1,000 RPM** |
| Electric fuel pump | **ON** |
| Primer | **LOCKED** |
| Flaps | **SET (0° normal / 25° short field)** |
| Trim | **SET for takeoff (neutral)** |
| Seat belts / harnesses | **FASTENED** |
| Door | **LATCHED (top and side)** |
| Pitot heat / lights | **AS REQUIRED** |
| Transponder | **ALT** |
| Takeoff briefing | **COMPLETE (abort point, EFATO plan)** |

### After Takeoff / Climb

| | |
|---|---|
| Airspeed | **79 KIAS (Vy) / 63 KIAS (Vx, obstacles)** |
| Flaps | **RETRACT (if used, when clear of obstacles)** |
| Electric fuel pump | **OFF (at safe altitude, check fuel pressure)** |
| Power | **FULL THROTTLE (climb)** |
| Mixture | **FULL RICH (lean above 5,000 ft)** |
| Engine instruments | **CHECK** |
| Landing light | **OFF (as desired)** |

### Cruise

| | |
|---|---|
| Power | **SET (2,300–2,600 RPM, 75% max)** |
| Mixture | **LEAN (per power chart)** |
| Fuel selector | **CHANGE TANKS every 30–60 min (pump ON during change)** |
| Engine instruments | **CHECK** |
| Fuel quantity | **CHECK against plan** |

### Descent

| | |
|---|---|
| Altimeter | **SET** |
| Mixture | **ENRICHEN as altitude decreases** |
| Carb heat | **AS REQUIRED (on for low power)** |
| Power | **REDUCE gradually (avoid shock cooling)** |
| Fuel selector | **FULLER TANK** |
| Approach briefing | **COMPLETE** |

### Before Landing

| | |
|---|---|
| Seat belts / harnesses | **FASTENED** |
| Fuel selector | **FULLER TANK** |
| Electric fuel pump | **ON** |
| Mixture | **FULL RICH** |
| Carb heat | **AS REQUIRED** |
| Flaps | **SET (below 103 KIAS; 40° normal landing)** |
| Airspeed | **63 KIAS final (flaps 40°)** |
| Landing light | **ON** |

### After Landing

| | |
|---|---|
| Flaps | **RETRACT** |
| Electric fuel pump | **OFF** |
| Carb heat | **OFF** |
| Transponder | **STBY / as required** |
| Lights | **AS REQUIRED** |

### Shutdown

| | |
|---|---|
| Parking brake | **SET** |
| Avionics master | **OFF** |
| Electrical switches | **OFF** |
| Throttle | **1,000 RPM** |
| Mixture | **IDLE CUT-OFF** |
| Magnetos | **OFF (after prop stops)** |
| Master switch | **OFF** |
| Fuel selector | **OFF (if parking overnight / as per operator)** |
| Control lock / chocks | **INSTALL** |

## Abnormal procedures

### Alternator Failure

Ammeter zero / discharge, ALT annunciator

| | |
|---|---|
| Ammeter | **CHECK (zero output)** |
| ALT circuit breaker | **CHECK, RESET once** |
| ALT switch | **OFF, then ON** |

**▶ Alternator restored?**

**Yes:**

| | |
|---|---|
| Ammeter | **MONITOR** |

**No:**

| | |
|---|---|
| ALT switch | **OFF** |
| Electrical load | **REDUCE to minimum (one radio, transponder)** |
| Battery | **MONITOR (approx 30 min endurance)** |
| Deferred: Alternator failed – on battery power, limited electrical endurance. Land as soon as practical; no electrical flap impact (manual flaps). | **NOTED** |

### Low Oil Pressure

Indication below 25 psi at idle / below 60 psi in flight

| | |
|---|---|
| Oil pressure / temperature | **CROSS-CHECK** |

**▶ Oil temperature rising with low pressure?**

**Yes – real:**

| | |
|---|---|
| Power | **REDUCE to minimum required** |
| Landing site | **SELECT, prepare for engine failure** |
| Deferred: Low oil pressure – land as soon as possible, expect engine failure. | **NOTED** |

**No – gauge suspect:**

| | |
|---|---|
| Flight | **CONTINUE to nearest suitable airport, monitor** |
| Deferred: Oil pressure indication unreliable – land as soon as practical. | **NOTED** |

### High Oil Temperature

Indication above 245°F / in the red

| | |
|---|---|
| Mixture | **ENRICHEN** |
| Power | **REDUCE** |
| Airspeed | **INCREASE (lower nose)** |
| Oil pressure | **CHECK** |

**▶ Temperature decreasing?**

**Yes:**

| | |
|---|---|
| Flight | **CONTINUE, monitor** |

**No:**

| | |
|---|---|
| Deferred: Engine overheat – land as soon as possible, expect possible engine failure. | **NOTED** |

## Rough Running / Carburetor Ice

RPM loss, rough engine

Carb heat ..... **ON (expect initial further roughness)**

Mixture ..... **ADJUST for smoothest running**

Electric fuel pump ..... **ON**

Fuel selector ..... **SWITCH TANKS**

Magnetos ..... **CHECK L / R / BOTH (operate on best if one is faulty)**

Primer ..... **LOCKED**

**▶ Engine smooth again?**

**Yes:**

Carb heat ..... **LEAVE ON if icing conditions**

Deferred: Carb icing conditions – carb heat on at reduced power, reduced performance. ..... **NOTED**

**No:**

Power ..... **REDUCE to smoothest setting**

Deferred: Rough-running engine – land as soon as possible; expect engine failure. ..... **NOTED**

## Low Fuel Pressure

Fuel pressure below 0.5 psi or fluctuating

Electric fuel pump ..... **ON**

Fuel selector ..... **SWITCH to fuller tank**

Mixture ..... **FULL RICH**

**▶ Pressure restored?**

**Yes:**

Electric fuel pump ..... **LEAVE ON**

Deferred: Electric fuel pump on continuously – engine-driven pump suspect. Land as soon as practical. ..... **NOTED**

**No:**

Fuel exhaustion ..... **CONSIDER**

Prepare for engine failure – see Engine Failure in Flight.

## Flap Handle Will Not Latch / Flaps Inoperative

Flap handle ..... **RETRY, hold then release latch slowly**

**▶ Flaps hold a setting?**

**Yes:**

Flaps ..... **USE AVAILABLE setting**

**No:**

Flaps ..... **UP (handle stowed)**

Approach ..... **70–75 KIAS, flaps-up landing**

Deferred: Flaps inoperative – flaps-up landing, longer landing distance. ..... **NOTED**

## Door Open in Flight

Airspeed ..... **SLOW to approx 89 KIAS**

Storm window ..... **OPEN (cabin vent / overhead vent closed)**

Door ..... **PULL CLOSED, latch side then top**

**▶ Door latched?**

**Yes:**

Flight ..... **CONTINUE**

**No:**

Aircraft is controllable with door unlatched. Expect noise and some yaw.

Deferred: Door unlatched – land normally, secure door on the ground. ..... **NOTED**

## Vacuum Failure

Vacuum below 4.8 in Hg / gyros drifting

Vacuum gauge ..... **CHECK**

Attitude indicator / DG ..... **CROSS-CHECK against turn coordinator and compass**

Attitude indicator / DG ..... **COVER or disregard**

IMC ..... **EXIT, request no-gyro assistance if needed**

Deferred: Vacuum failed – no attitude indicator / DG. Avoid IMC; partial-panel landing. ..... **NOTED**

## Static Source Blocked

Airspeed / altimeter / VSI erratic

Pitot heat ..... **ON**

Alternate static source ..... **OPEN (valve under left side of panel)**

Storm window / vents ..... **CLOSED (for calibration accuracy)**

Airspeed and altimeter read slightly high with alternate static; see POH correction table.

Deferred: Alternate static in use – small airspeed/altimeter errors on approach. ..... **NOTED**

### Inadvertent Icing

| | |
|---|---|
| Pitot heat | **ON** |
| Carb heat | **ON** |
| Cabin heat / defroster | **ON (windshield)** |
| Exit icing conditions | **TURN / CLIMB / DESCEND** |
| Airspeed | **INCREASE, avoid abrupt maneuvers** |
| Flaps | **LIMIT (flaps-up or 10° landing with residual ice)** |

Deferred: Residual airframe ice – flaps-up / 10° landing at increased speed, longer runway. **NOTED**

## Emergency procedures

### Engine Failure During Takeoff Roll

Loss of power before rotation

| | |
|---|---|
| Throttle (memory) | **CLOSED** |
| Brakes (memory) | **APPLY, stop straight ahead** |
| Mixture | **IDLE CUT-OFF** |
| Magnetos | **OFF** |
| Master switch | **OFF** |
| Fuel selector | **OFF** |

### Engine Failure After Takeoff

Airborne, insufficient runway remaining

**LAND IMMEDIATELY**

| | |
|---|---|
| Airspeed (memory) | **73 KIAS (best glide)** |
| Landing area (memory) | **SELECT straight ahead (small heading changes only)** |

**▶ Altitude and time available for a restart attempt?**

**Yes:**

| | |
|---|---|
| Fuel selector | **SWITCH to other tank** |
| Electric fuel pump | **CHECK ON** |
| Mixture | **FULL RICH** |
| Carb heat | **ON** |
| Primer | **LOCKED** |

**No – commit to landing:**

| | |
|---|---|
| Flaps | **AS REQUIRED** |
| Fuel selector | **OFF** |
| Mixture | **IDLE CUT-OFF** |
| Magnetos | **OFF** |
| Master switch | **OFF (before touchdown)** |
| Door | **UNLATCH before touchdown** |

Do not attempt to turn back to the runway at low altitude.

## Engine Failure in Flight

Loss of power with altitude available

**LAND AS SOON AS POSSIBLE**

| | |
|---|---|
| Airspeed (memory) | 73 KIAS (best glide) |
| Fuel selector (memory) | SWITCH to tank containing fuel |
| Electric fuel pump (memory) | ON |
| Mixture (memory) | FULL RICH |
| Carb heat (memory) | ON |
| Engine gauges | CHECK for cause (fuel pressure, oil, RPM) |
| Primer | LOCKED |

**▶ Propeller stopped (not windmilling)?**

**Yes – crank:**

| | |
|---|---|
| Magnetos | BOTH |
| Throttle | OPEN 1/4 inch |
| Starter | ENGAGE |

**No – windmilling:**

| | |
|---|---|
| Magnetos | CHECK BOTH (try L / R individually) |
| Throttle | CYCLE, then set as required |

**▶ Engine restarted?**

**Yes:**

| | |
|---|---|
| Power | REDUCE to minimum required |
| Engine instruments | MONITOR |
| Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect cause (fuel/carb ice). | NOTED |

**No – forced landing:**

Forced landing without power:

| | |
|---|---|
| Landing area | SELECT, plan approach into wind |
| Transponder / radio | 7700, MAYDAY (position, intentions) |
| Seat belts / harnesses | SECURE |
| Flaps | AS REQUIRED (full on final) |
| Airspeed | 63 KIAS final (flaps 40°) |
| Fuel selector | OFF |
| Mixture | IDLE CUT-OFF |
| Magnetos | OFF |
| Master switch | OFF (when no longer required) |
| Door | UNLATCH before touchdown |

## Engine Fire in Flight

**LAND IMMEDIATELY**

| | |
|---|---|
| Fuel selector (memory) | OFF |
| Throttle (memory) | CLOSED |
| Mixture (memory) | IDLE CUT-OFF |
| Electric fuel pump (memory) | CHECK OFF |
| Cabin heat / defroster (memory) | OFF |
| Airspeed | INCREASE to blow out fire (do not exceed 126 KIAS) |
| Forced landing | EXECUTE (do not attempt restart) |
| Master switch | OFF before touchdown |

## Engine Fire During Start

| | |
|---|---|
| Starter (memory) | CONTINUE CRANKING (draw fire into engine) |
| Mixture (memory) | IDLE CUT-OFF |
| Throttle (memory) | FULL OPEN |
| Electric fuel pump | OFF |
| Fuel selector | OFF |

**▶ Fire continues?**

**Yes:**

| | |
|---|---|
| Magnetos / master | OFF |
| Aircraft | EVACUATE, fire extinguisher |

**No:**

| | |
|---|---|
| Engine | SHUT DOWN, inspect before further flight |

## Electrical / Cabin Fire

**LAND AS SOON AS POSSIBLE**

- Master switch (memory) ..... **OFF**
- Avionics master (memory) ..... **OFF**
- Cabin heat / defroster (memory) ..... **OFF**
- Vents ..... **OPEN**
- Fire extinguisher ..... **USE (if fire in cabin)**

After discharge, ventilate the cabin before landing.

**▶ Fire out and electrical power essential?**

**Yes:**

- Circuit breakers ..... **CHECK, do not reset tripped breakers**
- Master switch ..... **ON**
- Electrical items ..... **ON one at a time, watch for smoke**
- Deferred: Electrical fire – faulted circuit isolated, affected equipment unavailable. No flaps impact (manual flaps). ..... **NOTED**

**No:**

Continue with master off. Manual flaps and engine are unaffected; no radio, lights or transponder.

## Emergency Descent

- Throttle (memory) ..... **CLOSED**
- Mixture (memory) ..... **FULL RICH**
- Carb heat (memory) ..... **ON**
- Flaps (memory) ..... **UP**
- Airspeed (memory) ..... **126 KIAS smooth air / 111 KIAS turbulent**

Reduce speed before extending flaps (103 KIAS max).

## Spin Recovery

Unintentional spin

- Throttle (memory) ..... **IDLE**
- Ailerons (memory) ..... **NEUTRAL**
- Rudder (memory) ..... **FULL OPPOSITE to rotation**
- Control wheel (memory) ..... **FULL FORWARD**
- Rudder (memory) ..... **NEUTRAL when rotation stops**
- Control wheel ..... **EASE BACK, recover from dive**

Intentional spins are prohibited.

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Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist version 1.8 (build c9e52546) · 2026-11-07